

Supplement for RDD results

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Abstract

Quarto document renders the outputs for the full set of RDD results (sensitivity check etc). Results are also saved in /paper-outputs as a .tex fragments so they can be pulled into overleaf directly

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1 Topline

- robust effect for authors a 3rd note (`have_another_note = note_counts[(note_counts >= 3)]`)
- with authors a 2nd note (`have_another_note = note_counts[(note_counts >= 2)]`)
- – robust rd not sig, donut RDD sig for 2 “larger holed donuts”, ordinary OLS is sig with 0.08 threshold,

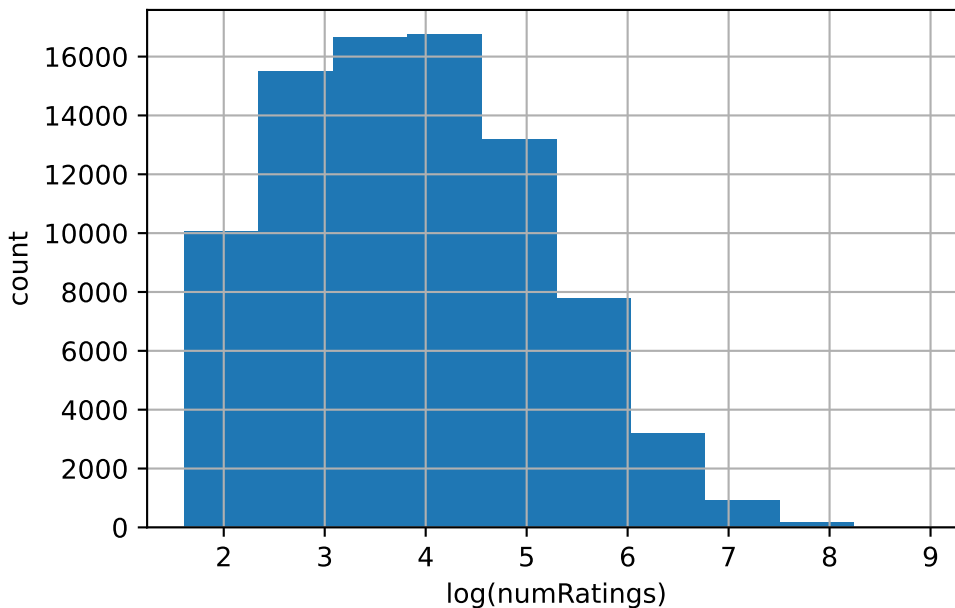
2 Setup & Data

```
Imported data for 3173276 notes
ALERT! re_auth_thresh = 2
Now we have 330234 first time note authors (all time)
Now we have 260997 first time note authors 2024-01-01 to 2026-06-02
Now we have 158099 notes written by the 4 major algorithms
Now we have 113635 notes classified as MISLEADING
Now we have 87252 notes which have enough ratings
Now we have 84253 aligned notes
--
FINAL total 84253 Notes
of which author writes again = 56996

NB Aligned data, current score
```

3 Plots

Most notes have a small number of rating, a small number have very many ratings This means the distribution of ratings is very skewed. We take the log of the number of ratings to make the distribution more normal



The basis for our RDD is the discontinuity between the intercept and note status (which defines publication / the possibility for publication)

Plotting $\log(\text{numRatings})$ vs NoteIntercept, with colours according to Note final status (finalRatingStatus)



4 Trad OLS RDD

Just using OLS and an arbitrary window

bandwidth = 0.08

Number of treated units: 5302

Number of control units: 5719

Estimated Treatment Effect at the Cutoff: 0.0476

I think we can interpret this as a % change the chance of authoring again if published

P-value for treatment: 0.0192

T-statistic for treatment: 2.3429

5 RDD using rdrobust

This has the option of using a polynomial fit (which we decline), and uses all points, weighted by proximity to cutoff

References for this method

@article{calonico2014robust, title = {Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {Econometrica}, volume = {82}, number = {6}, pages = {2295–2326}, year = {2014}, doi = {10.3982/ECTA11757} }

@article{calonico2018effect, title = {On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Farrell,

Max H.}, journal = {Journal of the American Statistical Association}, volume = {113}, number = {522}, pages = {767–779}, year = {2018}, doi = {10.1080/01621459.2017.1285776} }

@article{calonico2014rdrobust, title = {Robust Data-Driven Inference in the Regression-Discontinuity Design}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {The Stata Journal}, volume = {14}, number = {4}, pages = {909–946}, year = {2014}, doi = {10.1177/1536867X1401400413} }

running variable (x) length 84253

binary outcome (y) length 84253

cutoff (c) is 0.4

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	84253
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
-----	-----	-----
Number of Observations	74382	9871
Number of Unique Obs.	74305	9861
Number of Effective Obs.	3954	4284
Bandwidth Estimation	0.064	0.064
Bandwidth Bias	0.112	0.112
rho (h/b)	0.576	0.576

Method	Coef.	S.E.	z-stat	P> z	95% CI
-----	-----	-----	-----	-----	-----
Conventional	0.006	0.026	0.228	8.200e-01	[-0.045, 0.057]
Robust	-	-	-0.182	8.558e-01	[-0.064, 0.053]

MSE-optimal h = 0.0644

Bias-corrected point estimate: -0.0055 95% CI [-0.0644, 0.0534]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0322 (0.5x)	tau = -0.0247	95% CI [-0.1220, +0.0514]	N_L/N_R = 1352/2069
h = 0.0644 (1.0x)	tau = +0.0059	95% CI [-0.0644, +0.0534]	N_L/N_R = 3954/4284
h = 0.1289 (2.0x)	tau = +0.0451	95% CI [-0.0011, +0.0780]	N_L/N_R = 13215/8019

Mephi suggested text :

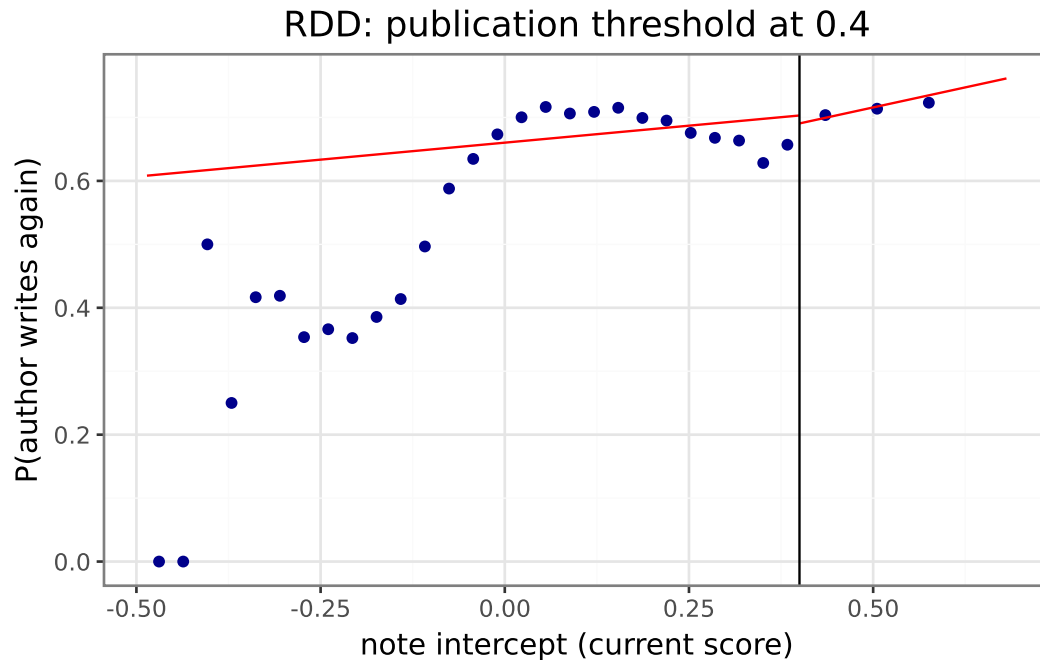
At the MSE-optimal bandwidth (h = 0.064) we estimate $\hat{\tau} = +0.6\text{pp}$ in the probability of writing a second note (conventional local-linear estimator; robust bias-corrected 95% CI [-6.4, +5.3], p = 0.86).

6 Converting rate to baseline outcome rate

Mephi suggested text: Untreated authors at the cutoff re-author at 63.9%; treated authors at the cutoff re-author at ~64.5%. Treatment effect = 0.6pp, a 0.9% relative lift over the counterfactual baseline.

7 Showing an illustrative RDD plot

First, binned means with gloabl linear fit (this is the default plot from `rdplot`)



```
['rdplot_mean_bin', 'rdplot_mean_x', 'rdplot_mean_y', 'rdplot_min_bin', 'rdplot_max_bin', 'rdplot_data']
paper-outputs/RDDrobust_scatter.png + paper-outputs/RDDrobust_scatter_data.csv written (30 bins)
```

RD plot at $c = 0.4$ ($p=1$, triangular, $N=84253$)

bins used : 27 left / 4 right

J_IMSE : [27.0, 4.0]

J_MV : [658.0, 565.0]

scale : [1, 1]

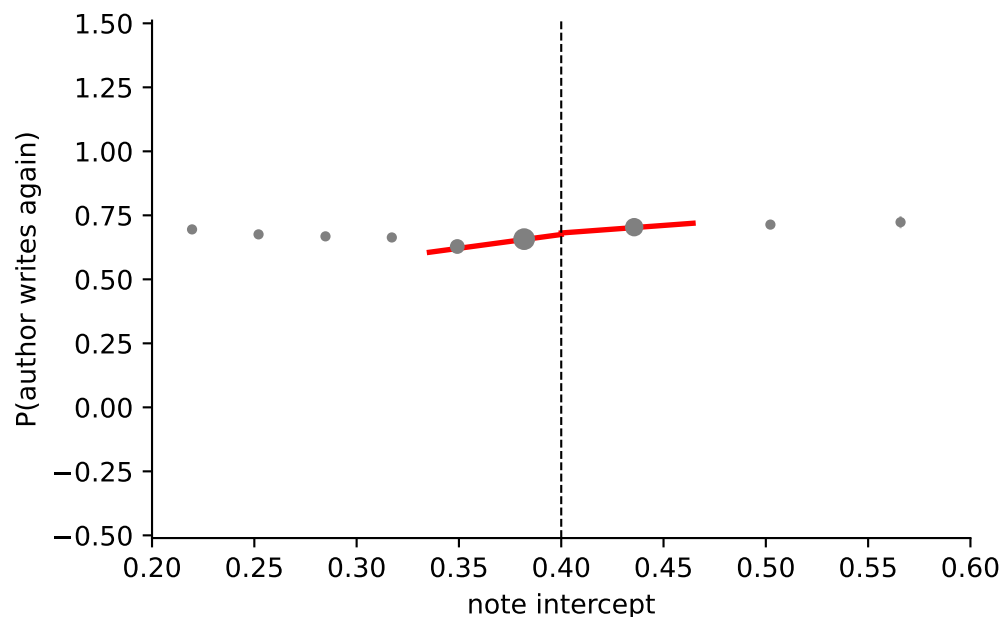
bin width : avg [0.032803518683822065, 0.07023575305938719], median [0.032803518683822086,

coefficients:

	Left	Right
0	0.703109	0.690572
1	0.107083	0.252484

wrote `paper-outputs/RDDrobust_scatter_meta.json`, `paper-outputs/RDDrobust_scatter_coef.csv`

Now with the local-linear fits (over the `rdrobust` defined window)



consistency check: gap at cutoff should conventional tau
 plot gap at c = +0.0059 vs conventional tau = +0.0059

8 Robustness check - density test

Next, manipulation test/density smoothness at cutoff

if units (people) can select to be just above cutoff then density will not be smooth (think the pass grade on an exam)

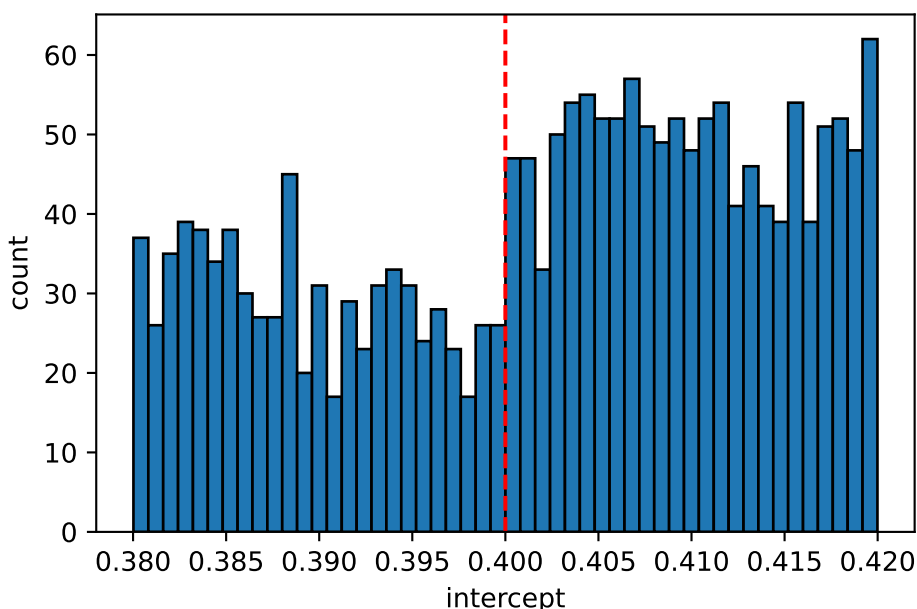
Inspecting density with a window of +/- 0.02

```
t_asy      NaN
t_jk      5.647629e+00
p_asy      NaN
p_jk      1.626760e-08
dtype: float64
left      0.371600
right     0.756969
diff      0.385369
dtype: float64
left      0.050657
right     0.041595
dtype: float64
full      84253
left      74382
right     9871
eff_left   2641
eff_right  2690
dtype: int64
leftN      [735.0, 877.0, 1067.0, 1256.0, 1428.0, 1629.0,...
```

```

rightN      [1226.0, 1397.0, 1541.0, 1706.0, 1880.0, 2053....
leftWindow  [0.02, 0.023406304522946866, 0.026812609045893...
rightWindow [0.02, 0.022399394497691624, 0.024798788995383...
pval        [9.633274573214021e-29, 8.163858561727352e-28,...
dtype: object
['X_max', 'X_min', 'binNW', 'bino', 'binoN', 'binoNStep', 'binoP', 'binoW', 'binoWStep', 'bino']

```



A wider window shows that the density does progressively decrease up to the cutoff, but no signs of sub-threshold notes being shifted to above threshold (decline not matched by leap post-cutoff)

9 Robustness check - Donut RDD

Excluding results at the cutoff

donut	n_eff_L	n_eff_R	h	coef_conv	se_conv	p_robust	ci_robust_lo	ci_robust_hi
none	3954	4284	0.064	0.0059	0.0260	0.8558	-0.0644	0.0534
±0.005	4607	4467	0.072	0.0330	0.0279	0.4510	-0.0405	0.0911
±0.010	5984	4981	0.085	0.0650	0.0277	0.0625	-0.0034	0.1333
±0.020	9277	5890	0.110	0.0638	0.0261	0.0897	-0.0092	0.1286

10 Sensitivity check - RDD sensitivity with different cutoffs

NB Only a cutoff of 0.4 makes sense in terms of the CN publication mechanism. Therefore we expect other thresholds to show now effect (definitely below 0.4, cutoffs above 0.4 will include some published notes above and below that cutoffs, so if there is a true effect may report an attenuated form of that)

```

running variable (x) length 84253
binary outcome (y) length 84253
cutoff (c) is 0.3

```

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 84253
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	65968	18285
Number of Unique Obs.	65907	18259
Number of Effective Obs.	10378	6390
Bandwidth Estimation	0.058	0.058
Bandwidth Bias	0.098	0.098
rho (h/b)	0.592	0.592

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.021	0.016	1.356	1.750e-01	[-0.01, 0.053]
Robust	-	-	1.392	1.639e-01	[-0.01, 0.062]

MSE-optimal h = 0.0578

Bias-corrected point estimate: 0.0256 95% CI [-0.0105, 0.0618]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0289 (0.5x) tau = +0.0362 95% CI [-0.0075, +0.0915] N_L/N_R = 4815/3751
h = 0.0578 (1.0x) tau = +0.0215 95% CI [-0.0105, +0.0618] N_L/N_R = 10378/6390
h = 0.1156 (2.0x) tau = +0.0106 95% CI [-0.0057, +0.0467] N_L/N_R = 23337/9356

running variable (x) length 84253

binary outcome (y) length 84253

cutoff (c) is 0.35

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 84253
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

Left	Right
------	-------


```

-----
Number of Observations      71782      12471
Number of Unique Obs.      71707      12459
Number of Effective Obs.    15725      6303
Bandwidth Estimation        0.106      0.106
Bandwidth Bias              0.166      0.166
rho (h/b)                  0.637      0.637

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional -0.018      0.016      -1.115      2.648e-01      [-0.049, 0.013]
Robust       -          -          -1.202      2.292e-01      [-0.059, 0.014]

```

MSE-optimal h = 0.1056

Bias-corrected point estimate: -0.0225 95% CI [-0.0592, 0.0142]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

```

h = 0.0528 (0.5x) tau = -0.0046 95% CI [-0.0499, +0.0560] N_L/N_R = 6271/2748
h = 0.1056 (1.0x) tau = -0.0178 95% CI [-0.0592, +0.0142] N_L/N_R = 15725/6303
h = 0.2112 (2.0x) tau = -0.0059 95% CI [-0.0403, +0.0157] N_L/N_R = 39676/11793

```

running variable (x) length 9871

binary outcome (y) length 9871

cutoff (c) is 0.45

Call: rdrobust

Sharp RD estimates using local polynomial regression.

```

Number of Observations:      9871
Polynomial Order Est. (p):      1
Polynomial Order Bias (q):      2
Kernel:                      Triangular
Bandwidth Selection:          mserrd
Var-Cov Estimator:          NN

```

```

-----
Left      Right
-----
Number of Observations      3285      6586
Number of Unique Obs.      3281      6580
Number of Effective Obs.    1108      1103
Bandwidth Estimation        0.016      0.016
Bandwidth Bias              0.024      0.024
rho (h/b)                  0.657      0.657

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional 0.071      0.041      1.705      8.827e-02      [-0.011, 0.152]
Robust       -          -          1.517      1.293e-01      [-0.022, 0.173]

```

MSE-optimal $h = 0.0160$

Bias-corrected point estimate: 0.0756 95% CI [-0.0221, 0.1733]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0080 (0.5x) tau = +0.1226 95% CI [+0.0050, +0.2791] N_L/N_R = 565/594
h = 0.0160 (1.0x) tau = +0.0706 95% CI [-0.0221, +0.1733] N_L/N_R = 1108/1103
h = 0.0319 (2.0x) tau = +0.0671 95% CI [+0.0123, +0.1513] N_L/N_R = 2187/2145

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 9871
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	3285	6586
Number of Unique Obs.	3281	6580
Number of Effective Obs.	1108	1103
Bandwidth Estimation	0.016	0.016
Bandwidth Bias	0.024	0.024
rho (h/b)	0.657	0.657

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.071	0.041	1.705	8.827e-02	[-0.011, 0.152]
Robust	-	-	1.517	1.293e-01	[-0.022, 0.173]

All these are non-significant

11 Sensitivity check - data for different time windows

PERIOD 1, from 2025-04-08 to 2026-06-01

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 87791 first time note authors 2025-04-08 to 2026-06-01

Now we have 54744 notes written by the 4 major algorithms

Now we have 41554 notes classified as MISLEADING

Now we have 28011 notes which have enough ratings

Now we have 27376 aligned notes

--

FINAL total 27376 Notes
of which author writes again = 15214

NB Aligned data, current score
running variable (x) length 27376
binary outcome (y) length 27376
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 27376
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	23660	3716
Number of Unique Obs.	23657	3713
Number of Effective Obs.	1508	1749
Bandwidth Estimation	0.062	0.062
Bandwidth Bias	0.1	0.1
rho (h/b)	0.614	0.614

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.024	0.044	0.556	5.783e-01	[-0.061, 0.11]
Robust	-	-	0.222	8.240e-01	[-0.091, 0.114]

MSE-optimal h = 0.0616

Bias-corrected point estimate: 0.0117 95% CI [-0.0911, 0.1144]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0308 (0.5x) tau = +0.0060 95% CI [-0.1619, +0.1593] N_L/N_R = 527/870
h = 0.0616 (1.0x) tau = +0.0243 95% CI [-0.0911, +0.1144] N_L/N_R = 1508/1749
h = 0.1232 (2.0x) tau = +0.0582 95% CI [-0.0165, +0.1184] N_L/N_R = 4718/3073

PERIOD 1, from 2024-01-01 to 2025-04-07

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 172844 first time note authors 2024-01-01 to 2025-04-07

Now we have 103127 notes written by the 4 major algorithms

Now we have 71914 notes classified as MISLEADING

Now we have 59118 notes which have enough ratings

Now we have 56757 aligned notes

--

FINAL total 56757 Notes
of which author writes again = 41726

NB Aligned data, current score
running variable (x) length 56757
binary outcome (y) length 56757
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 56757
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	50614	6143
Number of Unique Obs.	50570	6141
Number of Effective Obs.	3800	3279
Bandwidth Estimation	0.085	0.085
Bandwidth Bias	0.133	0.133
rho (h/b)	0.64	0.64

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.027	0.027	1.001	3.167e-01	[-0.026, 0.081]
Robust	-	-	0.534	5.936e-01	[-0.047, 0.081]

MSE-optimal h = 0.0849

Bias-corrected point estimate: 0.0174 95% CI [-0.0466, 0.0815]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0424 (0.5x)	tau = -0.0007	95% CI [-0.1010, +0.0856]	N_L/N_R = 1185/1556
h = 0.0849 (1.0x)	tau = +0.0273	95% CI [-0.0466, +0.0815]	N_L/N_R = 3800/3279
h = 0.1697 (2.0x)	tau = +0.0499	95% CI [-0.0005, +0.0866]	N_L/N_R = 13385/5775